

Frontend Interview Preparation Handbook

Senior & Semi-Senior Level — English Edition

1. JavaScript Core Concepts

Core Topics:

- Scope and Hoisting
- Closures and practical use cases
- Event Loop, Call Stack, Queue and Microtasks
- Promises and async/await
- 'this' keyword, bind/call/apply
- Destructuring, spread and rest operators

Interview Questions & Sample Answers:

- Can you explain how the event loop works in JavaScript?
The event loop manages asynchronous operations by continuously checking the call stack and task queue. When the stack is empty, queued callbacks are executed.
- What is a closure, and where would you use it?
A closure allows a function to access variables from its outer scope even after that scope has closed. Commonly used for data encapsulation or maintaining state.
- What's the difference between == and ===?
'==' performs type coercion before comparison, while '===' checks both value and type directly.

Personal Example:

2. Architecture, Modularization & Design Patterns

Core Topics:

- ES6 Modules (import/export)
- Common patterns: Observer, Singleton, Factory
- Separation of concerns and modular architecture
- SOLID principles overview

Interview Questions & Sample Answers:

- How do you structure a scalable frontend project?
I organize code by feature modules and shared utilities, keeping separation between UI components, logic, and services.
- What design patterns have you used?
I've applied modular patterns, reusable components, and occasionally the observer pattern for event-driven flows.
- How do you ensure code maintainability?
Through consistent naming, reusable logic, linting, and proper folder structure aligned with business domains.

Personal Example:

3. HTML & CSS Advanced

Core Topics:

- Accessibility (semantic HTML, ARIA attributes)
- Modern layouts (Flexbox, Grid, responsive design)
- Performance optimization (lazy loading, critical CSS)
- Animations and transitions (GPU-friendly properties)

Interview Questions & Sample Answers:

- How do you ensure accessibility in your projects?
I use semantic HTML, test with screen readers, and add ARIA roles when needed for dynamic components.
- What's the difference between Flexbox and Grid?
Flexbox is one-dimensional (row or column), while Grid is two-dimensional, managing both rows and columns simultaneously.
- How do you optimize CSS for performance?
By minimizing unused styles, deferring non-critical CSS, and reducing reflows with transform/opacity animations.

Personal Example:

4. Testing (Unit & E2E)

Core Topics:

- Testing pyramid: unit, integration, end-to-end
- Jest/Vitest: mocks, spies, assertions
- React Testing Library, Cypress basics
- Testable architecture: separation of logic

Interview Questions & Sample Answers:

- What's your approach to testing a React component?
I test both logic and user behavior, using React Testing Library for realistic rendering and Jest for isolated logic.
- How do you mock API calls?
I use Jest mocks or MSW to simulate network responses and ensure predictable test environments.
- What tools do you use for E2E testing?
Cypress for user flows and Playwright for cross-browser consistency.

Personal Example:

5. APIs, JSON, REST & GraphQL

Core Topics:

- REST fundamentals: methods, status codes, headers
- GraphQL basics: queries, mutations, subscriptions
- Error handling, retries and aborting requests
- Fetch API / Axios / AbortController

Interview Questions & Sample Answers:

- What's the difference between PUT and PATCH?
PUT replaces an entire resource, PATCH modifies specific fields.
- How would you handle API errors gracefully?
By centralizing error handling, providing user feedback, and retrying or falling back where applicable.
- What are the advantages of GraphQL over REST?
GraphQL lets clients request only the data they need, reducing over-fetching and number of requests.

Personal Example:

6. Agile Methodologies

Core Topics:

- Scrum vs. Kanban
- Roles: Product Owner, Scrum Master, Developer
- Ceremonies: daily, sprint planning, review, retrospective
- Estimations and handling requirement changes

Interview Questions & Sample Answers:

- How do you estimate your work in sprints?
We use story points based on complexity and team velocity.
- How do you handle changing requirements?
Through backlog refinement and communication with the Product Owner to reprioritize tasks.
- What's your role during sprint planning?
I contribute to task breakdown, estimation, and identify potential blockers early.

Personal Example:

7. Interview Practice & Soft Skills

Core Topics:

- STAR Method (Situation, Task, Action, Result)
- Elevator pitch (30-second professional summary)
- Behavioral questions practice
- Showcasing teamwork, adaptability and ownership

Interview Questions & Sample Answers:

- Tell me about a challenging project you worked on.
I describe the context, my specific actions, and measurable outcomes—following the STAR method.
- How do you keep your code clean?
By applying consistent conventions, regular refactoring, and code reviews.
- What do you do when you disagree with a technical decision?
I present data or prototypes to support my point, and remain open to team consensus.

Personal Example:

Quick Reference Cheat Sheet

Closure: Allows access to outer scope variables even after the outer function has finished.

Event Loop: Manages async tasks by moving them between the call stack and queue.

Design Pattern: Reusable solution for recurring architectural problems, e.g., factory or observer.

Testing: Focus on testing logic and user-critical flows, not just coverage.

Agile: Iterative workflow with sprints, retrospectives, and adaptable planning.